Mahbub Ali

Research proposal

Machine Learning-Based Predictive Analytics for Early Disease Risk Detection Using Electronic Health Records

Level: Master’s by Research. Field: Computer Science.

Prepared for your university and the supervisors reviewing this proposal. This is not a current enrolment.

Candidate background: Bachelor’s in Computer Science and Engineering, Green University of Bangladesh, CGPA 3.54. Software Developer, 3+ years.

Status: proposal in preparation. This document is not completed research, not a publication, and it does not report experiments or model results.

Background

Electronic health records collect diagnoses, medications, laboratory results, and encounter histories over time. That longitudinal record may contain signals of disease risk earlier than a single visit would show. Machine learning has been studied for this kind of prediction, but a useful model still depends on the outcome definition, the validation design, and whether a reviewer can see why a prediction was made.

What follows is the study I intend to pursue. The specific condition, outcome label, and prediction window are not fixed yet. They should be chosen after a literature review and after a suitable dataset is identified, and they should be locked before any model comparison begins.

The practical background is healthcare software, not a finished prediction study. That work includes HealthConnect, PrimeMIS, Sakura ENT Center, prescription writing, and MediScan Pro, grouped in part under the company name SakuraEHealth. A May 2025 field summary for MediScan Pro, in Dhaka, reports report time, ease of use, and cost for an offline USB endoscopy workflow. It does not report classifier accuracy, and it is not this electronic-health-record study.

Problem statement

Disease risk is often recognised only after symptoms are established, when the opportunity to act earlier may already be narrower. Electronic health records hold information that could support earlier risk stratification, but those records are uneven, sometimes incomplete, and always sensitive.

The problem this proposal addresses is how to compare standard predictive models for early disease risk detection on electronic health record data, and how to explain the predictions, without treating a research metric as a clinical decision.

Objectives

- Review published research on machine learning for early disease risk detection using electronic health records.
- Select one ethically accessible dataset and document its population, features, missingness, and known biases.
- Define the prediction outcome and time horizon, then prepare the data without leaking information into the test set.

- Train and compare logistic regression, random forest, XGBoost, and support vector machine models.
- Evaluate the models with accuracy, precision, recall, F1-score, and ROC-AUC.
- Apply SHAP to interpret feature contributions for the selected model or models.
- State the privacy, bias, and clinical-use limits of any finding.

Research questions

- To what extent can these models estimate early disease risk from the features available in the selected electronic health records?
- How do logistic regression, random forest, XGBoost, and support vector machines compare on accuracy, precision, recall, F1-score, and ROC-AUC?
- Which features contribute most to the predictions according to SHAP, and how should those explanations be read?
- What limits of data quality, bias, and privacy must qualify the findings?

Methodology

The study is planned in six stages. None of these stages has been carried out, and no dataset has been selected.

1. Literature review. Survey peer-reviewed work on electronic health record risk prediction, evaluation practice, and explainable AI in healthcare. Use that review to fix the clinical question, the outcome definition, and the prediction horizon before any model is trained.

2. Dataset selection. Use a public dataset with a clear licence, or a dataset available under institutional approval. Record who it represents and what it cannot support. No identifiable patient data is collected for this proposal.

3. Data preparation. Define the cohort and the outcome from the selected records. Inspect missing values, coding differences, and class balance before choosing how to handle them. Where records are longitudinal, split training, validation, and test data by patient so that one person’s later visits do not leak into training.

4. Model development. Train logistic regression, random forest, XGBoost, and a support vector machine as comparable candidates. Tune hyperparameters on validation data only. Reserve the test set for one final comparison.

5. Evaluation and explanation. Compare the models with accuracy, precision, recall, F1-score, and ROC-AUC. Discuss the decision threshold in relation to false negatives and false positives. Apply SHAP so feature contributions can be read alongside the scores.

6. Ethical boundary. Keep the work inside research evaluation. A later clinical use would need separate validation, clinical collaboration, and the appropriate regulatory review. This proposal is not a plan to deploy a diagnostic system.

Candidate algorithms

These are candidate algorithms for the comparison. They have not been trained for this project, and no ranking among them is claimed.

- Logistic regression. A transparent linear baseline for binary risk estimation.
- Random forest. An ensemble of decision trees that can represent non-linear relationships among mixed clinical features.
- XGBoost. Gradient-boosted trees, included because they are often strong on structured tables when validation is kept strict.

- Support vector machine. A margin-based classifier trained on the same prediction task, included for comparison.

Evaluation metrics

The planned metrics are accuracy, precision, recall, F1-score, and ROC-AUC. There are no scores to report, because the models have not been trained. A high score on one metric would not, by itself, show that a model is suitable for clinical use.

Explainable AI

SHAP (SHapley Additive exPlanations) is the explainability method planned for this study. The intention is to report global feature importance and to inspect local explanations for individual records.

SHAP has not been applied in this project. When explanations exist, they will be a way to scrutinise the model. They will not be described as proof of clinical cause, and they will not replace a clinician's judgement.

Expected outcomes

If the study is completed under supervision, the expected outputs are prospective. This proposal does not report them as achieved.

- A written comparison of the four candidate models on one defined early-risk task.
- A clear account of the dataset, including the population it represents and the limits of that data.
- A SHAP-based discussion of feature contributions for the model chosen for interpretation.
- An explicit statement of privacy, bias, and the boundary between research performance and clinical use.
- Thesis material suitable for a Master's by Research in Computer Science, directed toward healthcare AI.

Preparation already underway

- The proposal text on this page and on the portfolio site.
- Reading on logistic regression, random forests, gradient boosting, and support vector machines.
- Study of Python, Pandas, and NumPy for later tabular preparation. No analysis has been run.
- Study of SHAP and its limits. The method has not been applied.